Process Improvement Model Based on Six Sigma for Administrative Incident Management at a University

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ABSTRACT

Nowadays, users are more demanding with respect to the quality of service, changes in the environment are happening faster and faster and the presence of competition leads the managers of an organization to seek continuous improvement in the processes they perform in order to reduce time, avoid duplication of activities, optimize resources and increase user satisfaction. Therefore, this study seeks to develop a model based on Six Sigma that improves the administrative incident management process in a university. The type of research is applied with a pre-experimental experimental design. The instruments selected for data collection were the questionnaire and the observation form. As for the results, it was demonstrated that by implementing the Six Sigma-based model, the percentage of administrative incidents attended increased, the time of attention was reduced, the cost of attention was reduced and the percentage of satisfaction of the university's users was increased. In conclusion, a Six Sigma-based model was developed that improved the administrative incident management process, obtaining favorable results for the institution and its users.

CCS CONCEPTS

• **Information systems - Information systems applications - Decision support systems - Online analytical processing**;

KEYWORDS

Six sigma, incidents, processes, model, management

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1 INTRODUCTION

Undoubtedly there are currently several methodologies, methods, techniques and tools that can be applied to improve the processes of an organization; however, there is no modern, holistic and systemic model where the business model of the organization is considered as a starting point to develop the preliminary diagnosis to improve the processes, in response to this deficiency it is proposed to develop a new model.

Throughout the world, organizations are seeking to transform themselves by improving their processes in order to face the various changes that are occurring rapidly. In this sense [1] defined it as the effort made by organizations to use their resources efficiently. It is therefore important to keep in mind the objectives set, to evaluate whether the necessary resources are available and to know how to manage them appropriately [2]. In this respect [3] they also refer to the various changes that are taking place today, which have caused customer demands to increase with respect to the quality of service in all organizations, so they show greater interest in the processes they have and evaluate whether they are really efficient. They also mention that they must also pay attention to internal customers because they are the ones who carry out the processes and provide the service to users. In this sense [4] proposes that the success of a project is subject to two conditions: business value creation and management performance. The first is obtained as long as the achievements are aligned with the heart of the business and the second considers the objectives established from the beginning to the end.

Undoubtedly, in current times, institutions place a special value on service quality, because they are aware that if they manage to turn it into a competitive advantage, it will guarantee their competitiveness in the market. That is to say, user satisfaction is a great indicator to measure the efficiency of the processes, always remembering that the client is the main reason why an organization elaborates products to sell them or to provide a service. [5].

This is why universities are not oblivious to the changes and demands that exist in the world, because their objective is also focused on achieving excellence in their processes, optimizing them and guaranteeing their efficiency. In other words, the actions of the present will allow planning for a future in which the quality of attention will ensure the loyalty of users and thus the permanence of the institutions in the market [5].

On the other hand [6] the university is made up of a group of teachers and students where they learn and at the end of their studies they are awarded a degree as long as they comply with the established requirements. In other words, universities are educational institutions that provide a service to students, where the academic system is very important because it stores all the academic information (codes, grades, curricula, among others) and is one of the pillars of the institution. For this reason, it is required an agile management of all the data, to guarantee a quality in the

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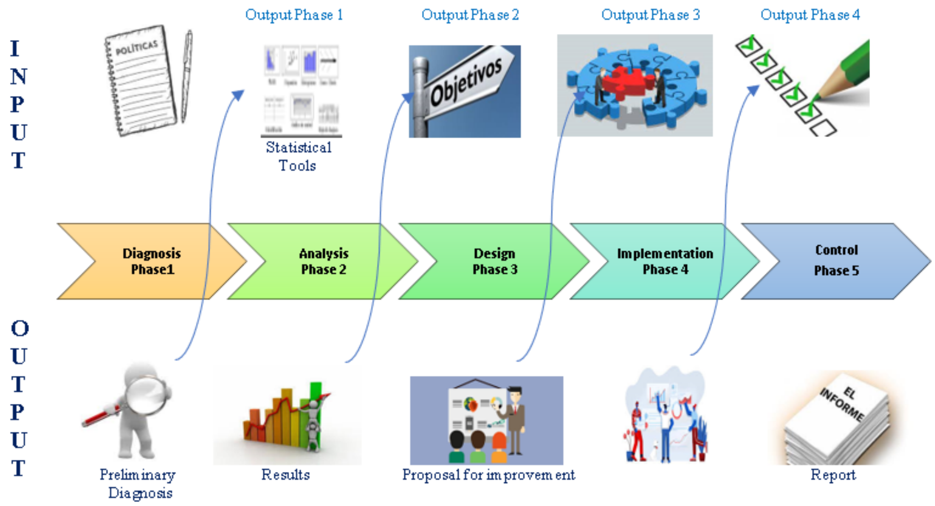

Figure 1: Context design.

service that is provided and with it to achieve that the user feels satisfied. [7].

2 LITERARURE REVIEW

According to [8] proposed the purpose of developing a procedure that, when applied by the organizations, would make it possible to constantly monitor the processes being executed in order to improve them. In this sense, he proposed a set of quantitative and qualitative techniques, to be used as deemed appropriate by the people responsible for applying the procedure and also take into account the actual situation. The aim is to identify the processes that require improvement, analyze what needs to be done and achieve continuous improvement.

For [9] proposed the implementation of five phases to achieve continuous improvement. That is to say, the methodology has five phases in which the reasons for the problem are investigated, then a proposal and an improvement plan are proposed, followed by implementation, monitoring and continuous follow-up in order to achieve an increase in productivity, lower costs, improved quality, customer satisfaction and greater synergy among workers.

According to [10] the objective was to design a set of activities aimed at achieving process improvement also considering the customer and the core business of the organization. Therefore it is necessary to take into account: the involvement of the personnel, the realization of a collaborative and cooperative work, contributions with creative and innovative ideas. The phases are: organize, identify the processes to be improved, represent the process, establish the improvement and test the established changes.

For [11] the objective was to establish the level of incidence of the continuous improvement model, which is based on the process that considers the quality of the service provided to the customers of the company where the research was carried out. As results it was observed that there is a difference between the data taken at the beginning and at the end of the application of the continuous improvement model, concluding that this model based on processes does have a positive impact on the level of service quality in the perception of the company's customers.

According to [12] proposed the creation of a methodology called MP-ISOWO, which is the result of the fusion of three methodologies Lean Six Sigma, Kaizen and BPM, covering the planning, execution, follow-up and pretest phases. The objective of his research was to seek an improvement in the process of incident management in the company. As a result, we obtained an increase in the KPI indexes, concluding that the application of statistical tools to analyze the data was very important to achieve the improvement of the incident management process.

3 METHODOLOGY

Nowadays there are several methodologies, methods, techniques and tools that can be applied to improve the processes of an organization; however, there is not a methodology that is modern, holistic and systemic where the business model of the organization is considered as a starting point to elaborate the preliminary diagnosis to improve the processes, in response to this lack, the design of a new process improvement model is proposed.

In the Diagnosis stage, the institution's policies and procedures must be analyzed in order to prepare a preliminary diagnosis. This result will be the input for the Analysis stage, where the statistical tool to be applied will be evaluated and the results obtained are the input for the Planning stage, where the objectives and strategies to be implemented to improve the processes will be established. Once it is approved, it is considered as input for the Implementation phase and then it closes with the Control phase.

3.1 Model Validation

The new model was validated by ten experts, See table 2.

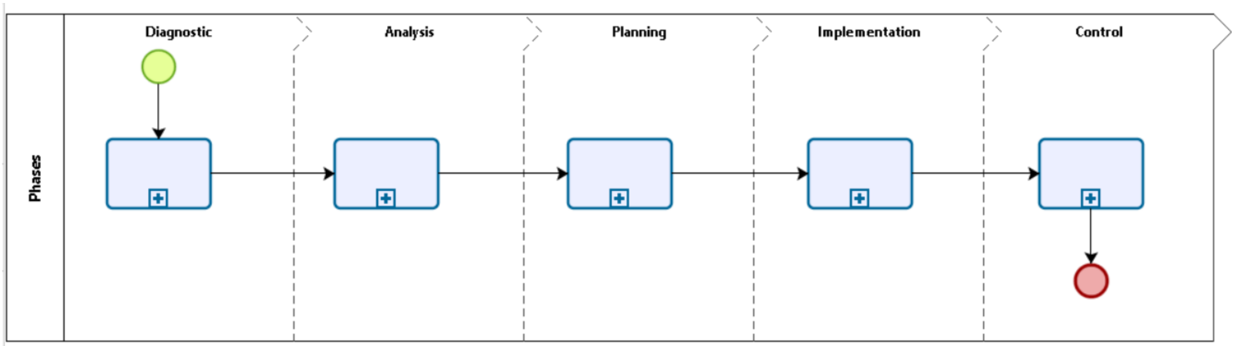

Figure 2: Flowchart at macro level.

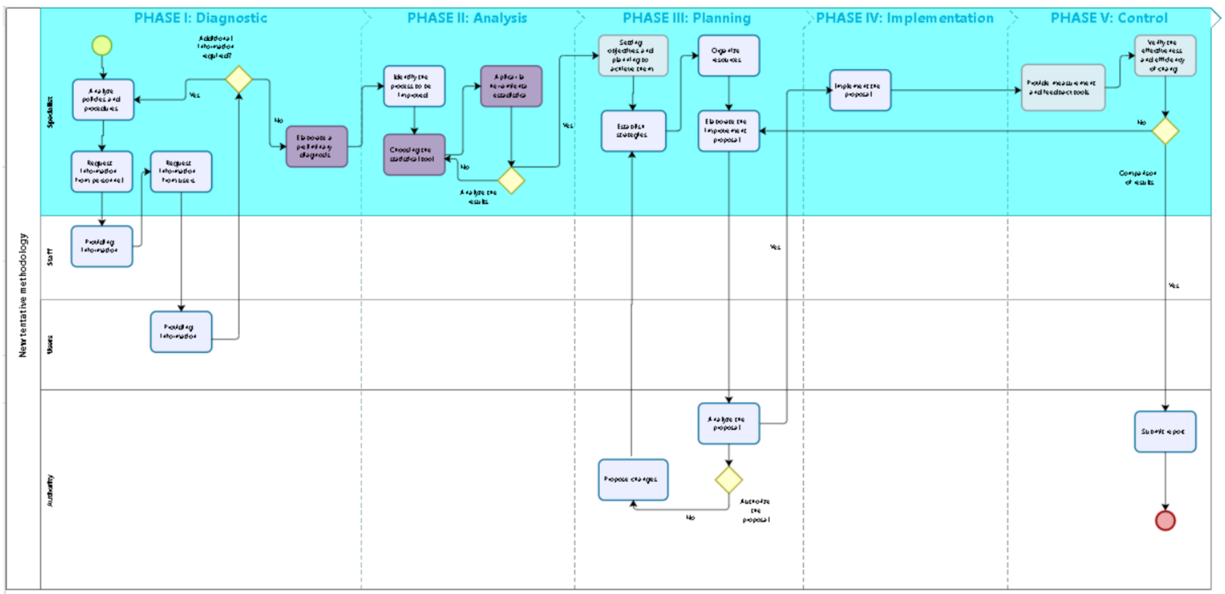

Figure 3: Flowchart at detailed level.

4 MODEL DEVELOPMENT

- Delays in the delivery of documents.
- Sometimes the documents issued contain some errors.
- Complex processes.
- User dissatisfaction.

Request information from users

The questionnaire was applied to determine the level of satisfaction with respect to the management of administrative incidents.

Elaborate the preliminary diagnosis

The following problems have been identified in the administrative incident management process:

- Low percentage of administrative incidents attended

Identify the process to be improved

After the interview with university personnel, it was determined that the process to be analyzed is the issuance of certificates.

Choosing the statistical tool

To analyze the causes of the problem identified, a cause-effect diagram was developed. See figure 4

The causes have been identified as corresponding to people, method, process and resources.

- In terms of personnel: the lack of experience of the staff has been identified, which contributes to inefficient management of administrative incidents. Also, there is little staff training and work overload in some months of the year when there is a demand for paperwork.
- Delay in the time it takes to attend to administrative incidents
- Increased cost of administrative incidents
- Low percentage of user satisfaction

Phase 2: Analysis

- In the method: there is evidence of a low pretest in the realization of the processes, on the other hand, there is no standardization of processes which is a big problem because

Table 1: Model Validation.

Phrases	**Activities**	Viable
Diagnosis	Analyze the organization's policies and resources.	Yes
	Request information from personnel	Yes
	Provide information	Yes
	Request information from users	Yes
	Provide information	Yes
	Elaborate a preliminary diagnosis	Yes
Analysis	Identify the process to be improved	Yes
	Choose the statistical tool	Yes
	Apply the statistical tool	Yes
	Analyze the results	Yes
Planificación	Setting objectives and planning to achieve them	Yes
	Establish strategies	Yes
	Organize resources	Yes
	Elaborate the improvement proposal	Yes
	Analyze the proposal	Yes
	Authorize the proposal	Yes
	Propose changes	Yes
Implementation	Implement the proposal	Yes
Control	Provide measurement and feedback tools	Yes
	Verify effectiveness and efficiency of changes	Yes
	Benchmarking results	Yes
	Reporting	Yes

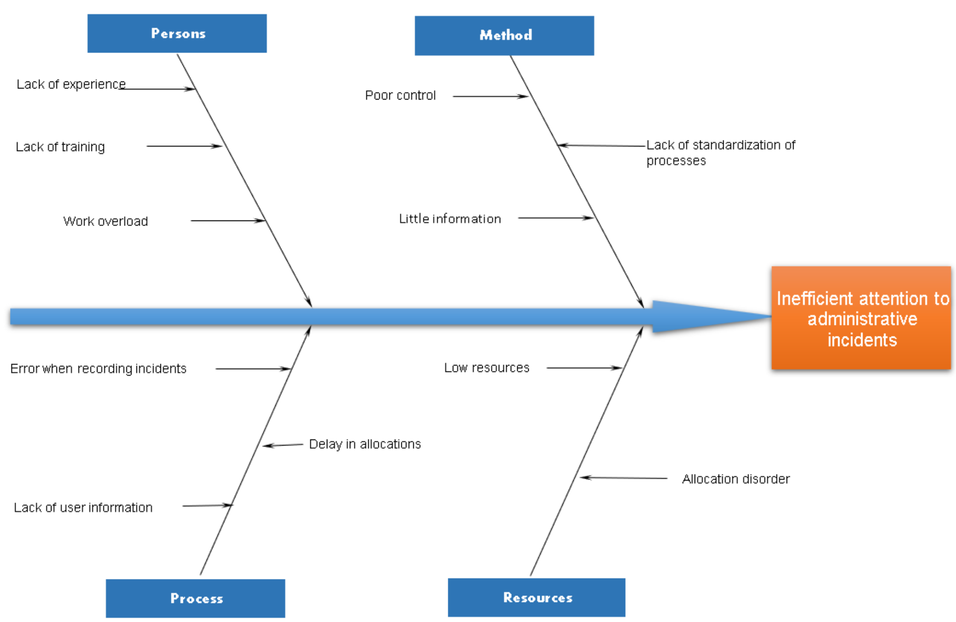

Figure 4: Cause-effect diagram.

the employee has his own criteria when trying to solve an administrative incident and finally the scarce information given at the time of induction and/or development of activities

- Process: there is no adequate record of the administrative incidents presented and the user is not provided with the necessary information to carry out the necessary procedures.

Table 2: Indicators.

INDICATORS	Average Values
• -Percentage of administrative incidents attended to per day	• 48.61%
• -Cost of administrative incidents attended to per day	• S/ 125.01
• -Time of administrative incidents handled per day	• 540 min
• -Percentage of user satisfaction	• 74.50%

Table 3: Matrix RACI.

Activities	User	Assistant	Boss
Identification	*C*		
Registration	*C*	*A*	
Research	*C*	*A*	
Solution	*I*	*A*	*R*
Closing	*C*	*A/I*	

- Resources: Poor order in the assignment to personnel of the incidents to be resolved and lack of the necessary resources and complete information to attend to reported cases.

Phase 3: Planning

Setting objectives

Once the causes for the inefficient handling of administrative incidents in the chosen process have been identified, it is necessary to establish the following objectives:

- Increase the percentage of administrative incidents handled per day.
- Reduce the time required to attend to administrative incidents per day.
- Reduce the cost of attending to administrative incidents per day.
- Increase the percentage of user satisfaction per day.

Establish strategies

A training and communication plan was designed to ensure that the staff is aware of the university's regulations, directives and provisions.

In terms of communication channels, corporate email accounts, meetings and documents are considered.

The following are the actions for developing team work.

Organizing resources

The RACI matrix specifies the following roles:

I: informed, C: consulted, A: in charge, R: responsible

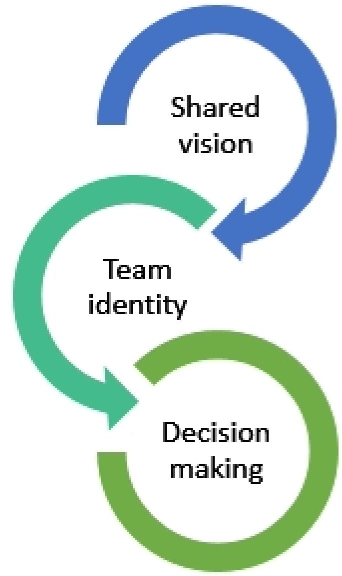

Figure 5: Actions to be developed.

Roles and Responsibilities

Head of the area:

- Provide guidance to staff on processes.
- Ensure that administrative incidents are solved.
- Manage a Good relationship with users.

Assistant:

- Record reported administrative incidents.
- Analyze the complexity of the incident..
- Find a solution in the shortest possible time.
- Keep the user informed.
- Document the solution provided.

Requirements

For the development of this research, the following was requested:

- Document each of the processes considering the provisions of the institution.

Phase 4: Implementation

The improvements proposed were implemented after obtaining approval.

- The personnel involved in the process were trained.
- Pre-test and post-test group times were taken.
- The results obtained were analyzed.

Phase 5: Control

After the training of personnel and the improvements proposed, the following results were obtained:

5 RESULTS

Of the sample under study, 63.3% of the incidents attended for both groups, while the total number of incidents assigned to the pretest group remained at 3 incidents, while for the posttest group it increased to 4 incidents.

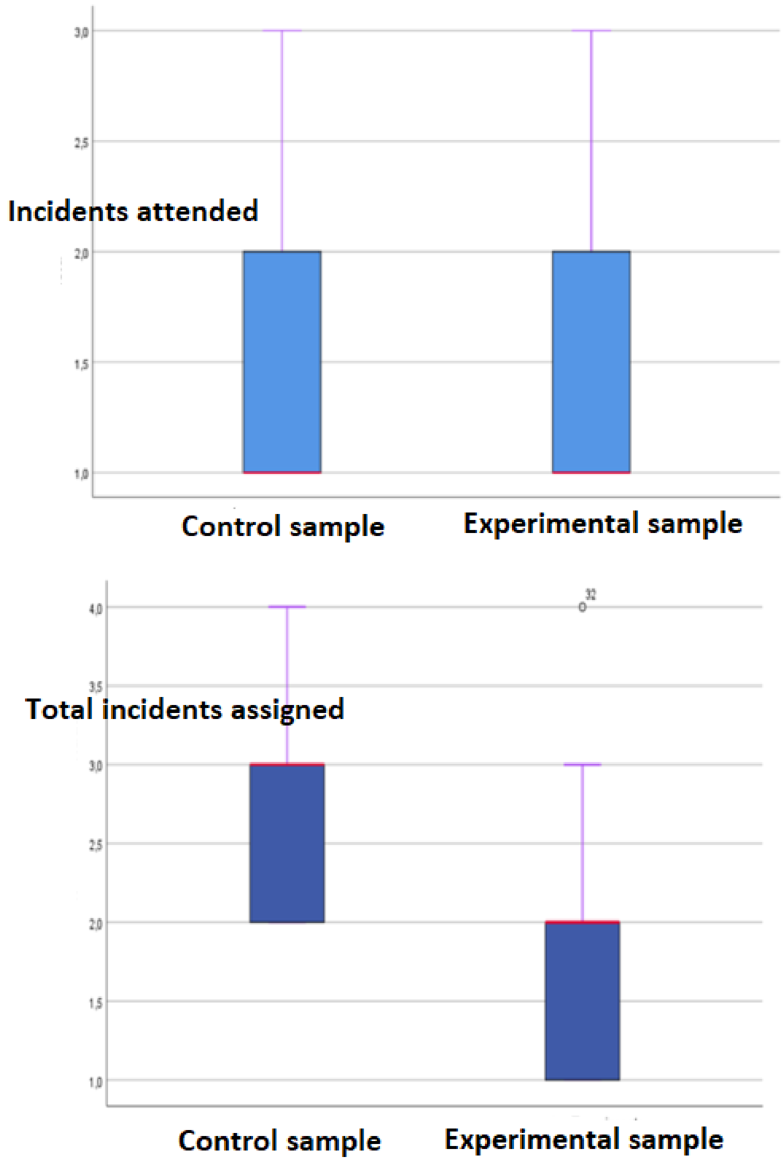

Figure 6: Percentage of administrative incidents attended to in academic records of a private university, applying a model based on Six Sigma.

Table 4: Indicators Results.

Indicators	Postest	Pretest
Percentage of administrative incidents handled per day	48.61%	80%
Cost of administrative incidents handled per day	S/ 125	S/ 26
Time of administrative incidents handled per day	540	111
Percentage of user satisfaction per day	75%	91%

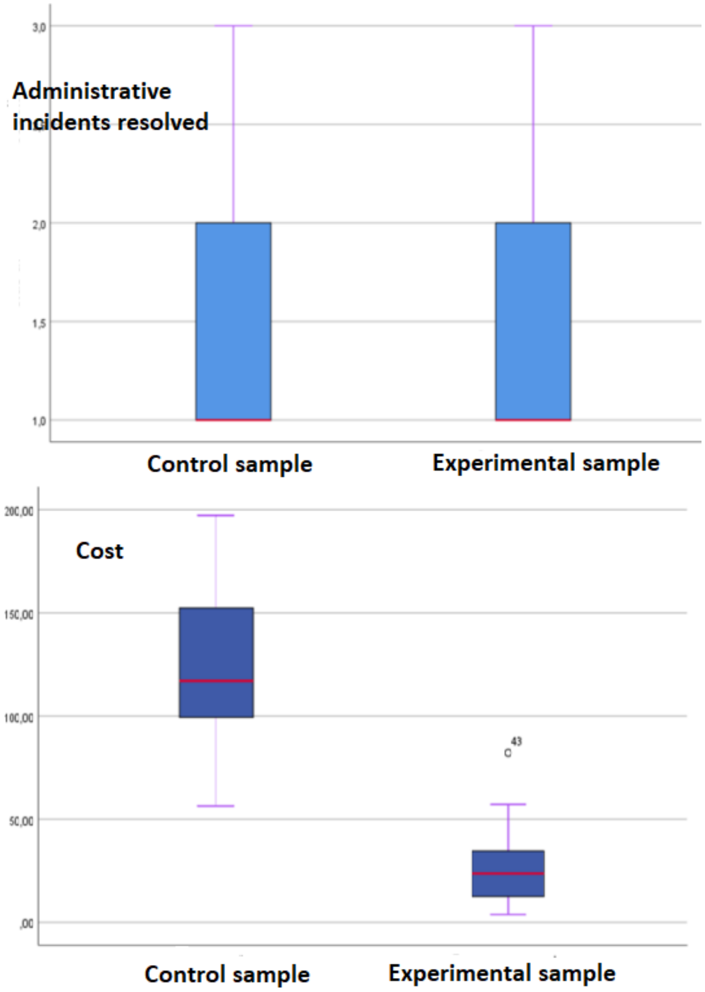

Figure 7: Cost of attention of administrative incidents in academic records of a private university, applying a model based on Six Sigma.

The average number of incidents attended in the pretest group is 1.4 incidents and for the posttest group 1.43 incidents, for the total incidents assigned in the pretest group is 2.90 incidents while the posttest group is 1.90 incidents out of 4 assigned.

From the pretest 63.3% (19) administrative incidents resolved with a cost for 19 incidents of 2519.7 soles, for the posttest sample 66.7% (20) administrative incidents resolved with a cost for 20 incidents of 585.9 soles.

The average number of administrative incidents resolved in the pretest group is 1.4 (± 0.56) incidents and for the posttest group 1.37 (± 0.55) incidents, the average cost of incidents in the pretest is 125.00 (± 38.66) soles and for the posttest is 25.58 (± 17.77) soles. 50% of the incidence cost for the pretest group is below 117.01 soles and for the posttest group below 23.66 soles.

Of the pretest sample 63.3% (19) administrative incidents resolved with an attention time for 19 incidents of 11,310 minutes, for the posttest sample 66.7% (20) administrative incidents resolved with an attention time of 2,531 minutes.

The average number of administrative incidents resolved in the pretest is 1.4 (± 0.56) incidents and for the posttest 1.37 (± 0.55) incidents, the average time of attention to incidents in the pretest is 716.43 (± 247.39) minutes and for the posttest is 138.03 (± 122.50) minutes. 50% of the incident attention time for the pretest group is below 718.5 minutes and for the posttest group below 122.50 minutes.

From the pretest 50% (15) were dissatisfied and 50% (15) were satisfied, for the posttest 100% (30) were satisfied.

6 DISCUSSION

In reference to the average attention to administrative incidents in the pretest group was 1.43 incidents and in the posttest group it was 1.90, evidencing an improvement. In addition, the percentage in the pretest group was 49% and 80% in the experimental group. As for the significance value, it was obtained that p-value =0.000 is less than α =0.05, concluding that the implementation of a model based on Six Sigma did increase the percentage of administrative

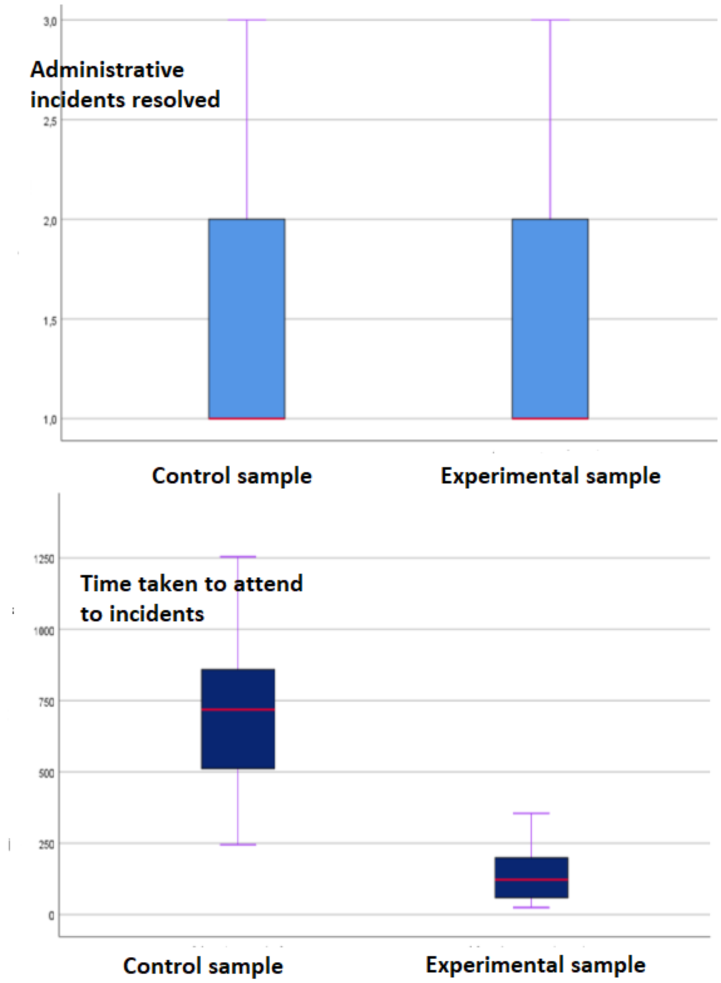

Figure 8: Reduction of administrative incident response time at a private university, applying a Six Sigma-based model.

incidents attended and total incidents assigned in academic records of a private university. The result obtained is similar to that obtained by Proaño et al. (2017) who determined that it is possible to improve the product, service or process in a company in order to remain in a competitive environment, in addition to worrying about achieving what was planned.

In reference to the attention time the average in the pretest group was 1.4 and 716.43 minutes, in the posttest group it was 1.37 and 138.03. In addition, it was shown that 50% of the incident attention time of the pretest group is below 718.5 minutes and for the experimental group below 122.50 minutes remaining as evidence that an improvement. As for the significance value, the p-value =0.000 was obtained, which is less than 0.005, concluding that the implementation of a model based on Six Sigma did reduce the attention time of the administrative incidents attended to in academic records of a private university. The result obtained is similar to that obtained by Alarcón (2017) where he showed that the continuous improvement model based on processes did have a significant impact on the response capacity, which was perceived and valued by customers.

In reference to the cost of attention to administrative incidents in the pretest group there was an average of 1.4 incidents with a cost of S/. 125.00, in the posttest group there was an average of 1.37 with a cost of S/. 2558. In addition, it was demonstrated that 50% of the cost of incidences of the pretest group is below 117.01 soles and for the experimental group below 23.66 soles, which is evidence of an improvement. As for the significance value, the p-value =0.000 was obtained, which is less than 0.005, concluding that the implementation of a model based on Six Sigma did reduce the cost of attention of the administrative incidents attended to in the academic records of a private university. The result obtained is similar to that obtained by Perez (2016) who indicated that it is very important to monitor the processes, involving an evaluation of everything necessary to improve and reduce costs.

In reference to user satisfaction in the pretest group it was found that 74.50% satisfaction and in the posttest group 90.67%. As for the significance value, the p-value =0.000 was obtained, which is

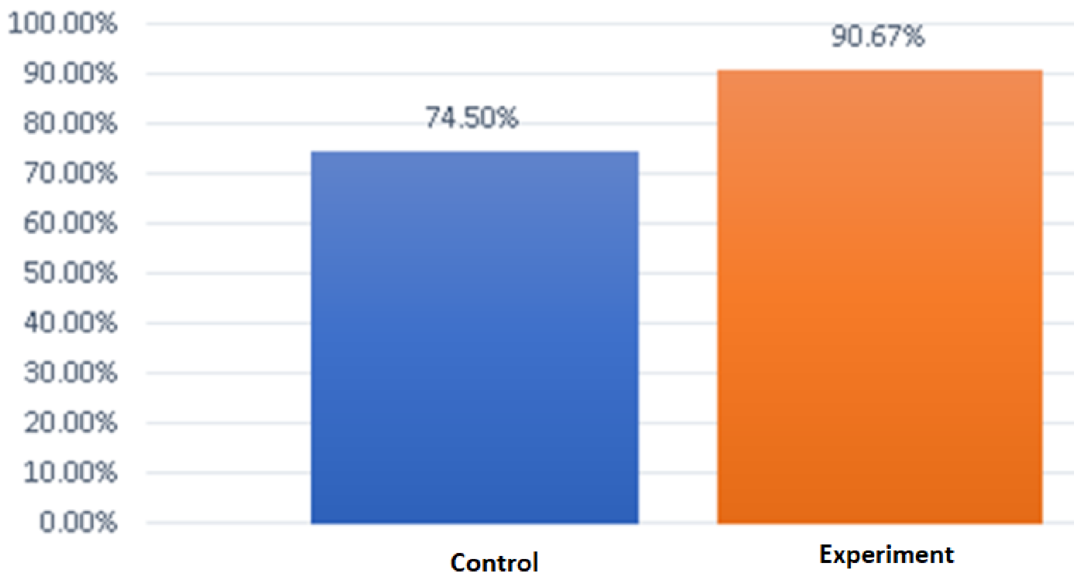

Figure 9: Percentage of user satisfaction in academic records of a university applying a model based on Six Sigma.

less than 0.005, concluding that the implementation of a model based on Six Sigma increased the percentage of user satisfaction in academic records of a private university. The result obtained is similar to that obtained by Delgado (2021) where he showed that with the implementation of the MP-ISOWO methodology, customer satisfaction increased from 87.4% to 94.57% in the company where the research was carried out.

7 CONCLUSIONS

A model based on Six Sigma was developed and implemented to improve the administrative incident management process in academic records of a private university. Likewise, with the implementation of a process improvement model based on Six Sigma, the percentage of incidents attended in the experimental group increased to an average of 80%; in the pretest group, the average was 49% in academic records of a private university.

In addition, the average time of attention was reduced to 4141 minutes in the experimental group; in the pretest group, the average time of attention was 21493 minutes for the total number of administrative incidents attended to in academic records of a private university. The average time of attention was 138.03 in the experimental group and 716.43 in the pretest group, and the cost of attention was also reduced, averaging S/. 767.6 in the experimental group and S/. 3651.8 in the pretest group. As for the average time of attention in the experimental group was S/. 25.58, in the pretest group it was S/. 125.00 of the total of administrative incidents attended to in academic records of a private university.

Finally, with the implementation of an improvement model based on Six Sigma, the percentage of user satisfaction increased in the experimental group by 90.67%, in the pretest group it was 74.50% of the total number of administrative incidents handled in academic records of a private university, and in the pretest group it was 74.50% of the total number of administrative incidents handled in academic records of a private university.

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